

What does that significance measure?

Replicating Section 4 of Homola et al. (2023)

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September 2026

What we set out to do

The claim

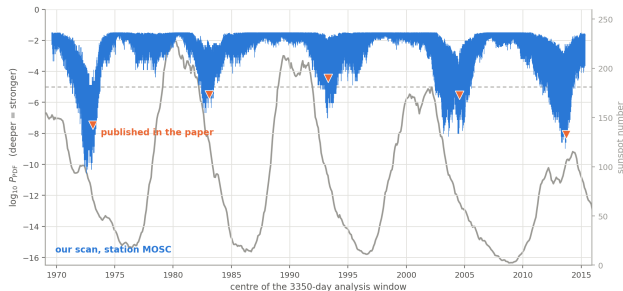
Homola et al., *JASTP* **247** (2023) 106068: a correlation between cosmic-ray flux and global seismicity that is **precursory** — earthquakes follow changes in CR at $\Delta t \approx 15$ d — with a **periodicity close to the solar cycle**. Reported: 6.23σ .

Our task

Reproduce the Section 4 recipe on our own data, then ask **what that significance actually measures**.

Data. NMDB neutron monitors (20 stations, 1965–2015, 6 h) · USGS catalogue $M \geq 4.0$ (507 919 events, 1965–2025) · SILSO sunspot number.

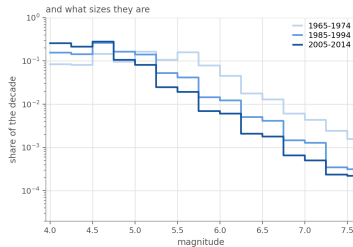
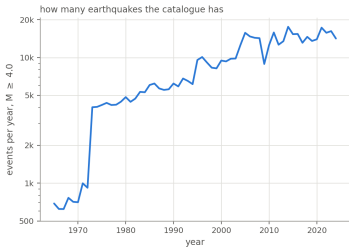
The recipe reproduces — and more stations will not help



- All five published values recovered at the paper's own dates, to 0.24–0.77 decades.
- Our four eras compose to 6.47 σ ; the paper's four give 6.23 σ .
- Every era found independently by 7 / 6 / 10 / 13 / 16 of the 20 stations.

20260906a, 20260906j, 20260906m

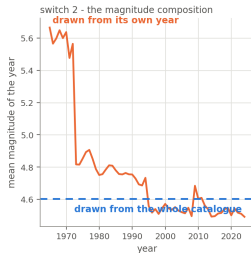
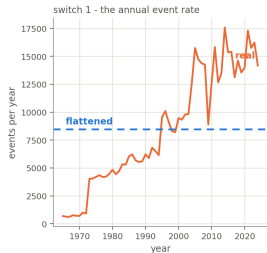
How much of this catalogue is the Earth?



- About 700 events in 1968, over 15 000 after 2010 — same magnitude threshold.
- Almost all of it small events: mean magnitude 4.90 in the 1970s, 4.55 in the 2000s.
- Plate tectonics did not change; the seismometer network did. Is that alone enough to produce the eras?

20260903c, 20260906m

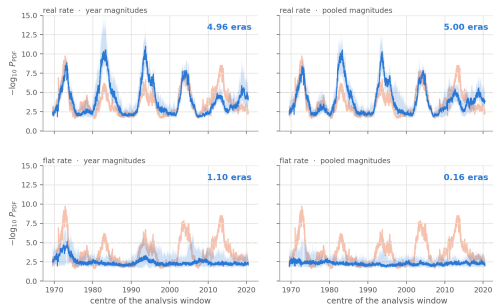
How we test whether that alone is enough



- Synthetic catalogues with **every event time redrawn uniformly inside its own year**: no structure **shorter than a year** survives. Event count identical: 507 919.
- **Two switches**: the annual event rate (real or flat), the magnitude randomisation (own year or global). Four combinations $\times$ 50 realizations, same methodology as on the real data.

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What the synthetic catalogues produce



Eras per realization (real: 5)

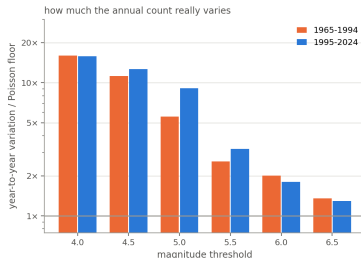
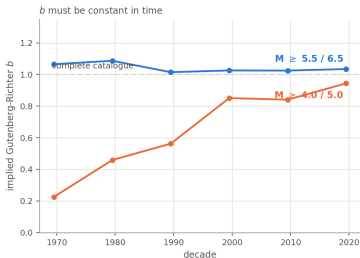
	magnitudes drawn from	
	own year	pooled
rate		
real	4.96	5.00
flat	1.10	0.16

The eras do not need the effect in order to appear.

The one the surrogate does *not* reproduce (6 of 50) is the last one — the era our completeness test flags as the **least contaminated**. The other four have the shape the catalogue alone produces.

20260906d, 20260906e, 20260906f

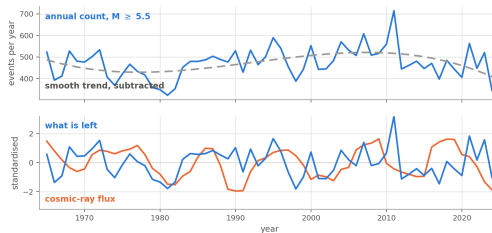
That multi-year shape is the instrument



- Gutenberg–Richter b must be **constant in time** if the catalogue is complete. It drifts **0.23** $\rightarrow$ **0.94** for $M \geq 4.0/5.0$, but holds 1.01–1.09 for $M \geq 5.5/6.5$.
- An annual count varies from year to year by $\sqrt{N}$ even if earthquakes were independent. Observed: **16x** that floor at $M \geq 4.0$, **1.3x** at $M \geq 6.5$ — yet aftershocks come in every size.

20260902d, 20260906m

Asking the claim where the catalogue is stable



- A deliberate departure from the paper's $M \geq 4.0$: $M \geq 5.5$ is the lowest complete threshold.
- Why detrend: both series drift over six decades, and a correlation on the drift says only that both went somewhere.
- Best p after Bonferroni, 3 methods $\times$ 6 thresholds: 0.42–1.00. The curves do not line up.

What we could not break

Two ways the eras might have been artefacts of the *method* rather than of the data. Both objections were ours; both failed.

Does the scan find eras at a fixed distance from where it starts?

Repeat the search on sub-ranges starting 3, 6 and 9 years later. An era pinned to the start of the scanned range would move *with* it, year for year; a real one would not move at all. Measured: an era date follows only 0.17 of the shift — rejection threshold 1/3.

Do the discarded bins break the $p = 0.5$ null hypothesis?

Bins lost to data gaps are not a random sample, so the surviving fraction could sit off 0.5 and inflate every era. Measured over the whole grid: 0.500, largest single departure 0.0045.

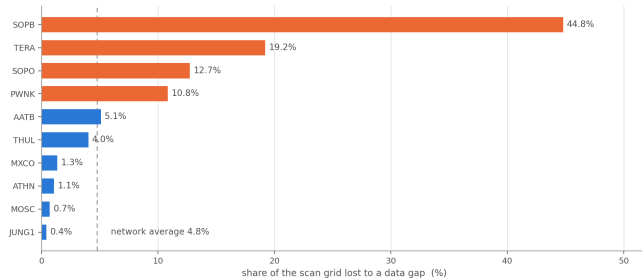
1. **We have not measured the method's sensitivity.** We can say the eras arise without any physical effect; we *cannot* say a weak real effect would have been visible, since it would be buried in the same trend artefact.
2. **Repeat the whole chain on a second catalogue.** GCMT is homogeneous since 1976; every step above runs on it unchanged.
3. **The result depends on the exact starting date.** Sliding t_0 by less than one 5-day bin changes P_{PDF} by about **four orders of magnitude** — the same era read a month apart gives 2.29 and 7.84. A significance quoted at one date therefore carries a spread that never gets reported. **Quoting the range over that bin** would fix it, and costs nothing.

Thank you

all code used can be found in this repository

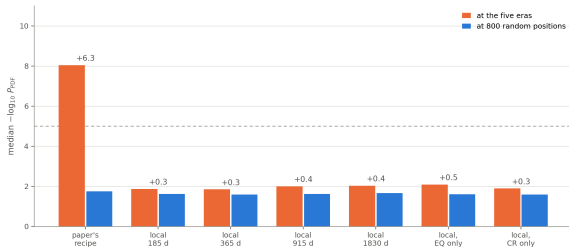
`github.com/mflagga/prak`

Additional: data quality across the station network



- **Checked:** completeness, missing data encoded as a real value, repeated readings, gaps that move the analysis window.
- **Excluded:** MCRL (117 readings of exactly 0.0), NANM (128 identical readings over 31.75 d).
- **Treat with care:** SOPB loses **44.8%** of its scan, three more over 10%; network average 4.8%.

Additional: the running-median variant



Normalising the two components by a running median over ± 40 bins (405 days) instead of by the median of the whole 3350-day window.

Era-to-background contrast collapses from +6.29 to +0.25/ +0.25/ +0.39 for windows of 205/405/1005 days.

Honest reading: the effect exists relative to a ~ 9 -year baseline and is absent relative to any baseline shorter than ~ 3 years.

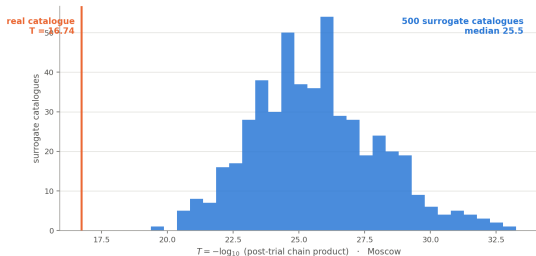
A 405-day running median also removes any ~ 11 -year modulation — the very thing the paper is about. So this result fixes the time scale the effect lives on, and says nothing about whether it is real.

Additional: measured against the paper's published counts

published point ($P = 1675$ d, at the paper's own t_0)	paper N_+	ours N_+	paper σ	ours σ
Moscow, recipe unchanged	199	194	3.5	3.02
Moscow, t_0 tuned over 13 400 positions	218	214	5.76	5.25
Oulu, t_0 tuned over 13 400 positions	220	217	5.92	5.39

- The earthquake side is verified on its own: for $M(S_m, P)$ — a quantity with no cosmic-ray measurement in it — the paper publishes 859.55 and we get 867.80.
- Our best Moscow point over the same scan reaches 5.92 σ , above the published 5.76 — but 35 days away from the published date.
- **Composing the four eras:** a product of p-values shrinks with every era added (four null results at $p = 0.5$ give 0.0625). By Fisher's method the same four numbers give 4.90 σ , not 6.23.

Additional: calibrating the significance on 500 surrogates

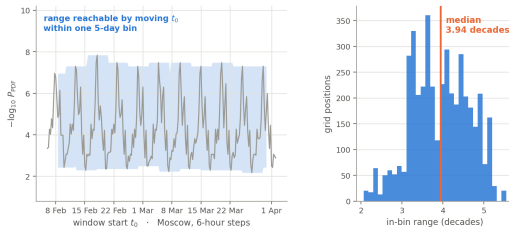


Every surrogate catalogue produces a **stronger** chain than the real one — **0 of 500**, for each of three stations, while the same chains read nominally give $12.18/8.67/8.42 \sigma$.

Caveat that must travel with this number

This does *not* mean the data are anti-correlated. The null distribution is **not variance-matched**: drawing times uniformly within the year destroys aftershock clustering, the noise that dilutes sign agreement — so the test is conservative in the wrong direction. *Pre-6 Sep 2026 binning convention; deliberately not recomputed.*

Additional: what a single significance number rests on



- The curve oscillates with a period equal to the bin width: moving t_0 inside *one* 5-day bin moves the result by **3.94 decades** (median).
- A significance quoted at one date is one draw from that band: our own per-station chains moved by **1.3–3.3 σ** when the era reference dates shifted by a month, while the curves did not deepen.
- The paper's Fig. 2 caption describes tuning t_0 with 20 small shifts across this range.

Additional: method parameters and scope

The statistic

- $P = 3350$ d, $d = 5$ d, 671 bins / 670 pairs; $\Delta t = 15$ d as in the paper, but the lattice is anchored on data, so the effective offset is 14.79 d
- earthquake term: magnitude sum over the window median; CR term: the CR change, normalised the same way; count bins where the two have the same sign, test against a fair coin
- grid: 6 h step, 940 792 positions; hygiene: drop a position when a gap relocates the window by more than one bin (4.8%)
- detection: height 5.0, separation 3350 d — so eras closer than 9.17 y cannot be returned, hence their spacing is not evidence of a solar rhythm

Scope of what was shown

- the surrogate test and the two null tests: one station, one catalogue
- the surrogate preserves the annual event count exactly, hence everything on multi-year scales — including any hypothetical ~ 11 -year modulation. It is a null for sub-annual coupling only; that is why the completeness and correlation results use none of this machinery
- two co-located monitors agree at 0.947 in raw counts, 0.26 in the derived curves
- cosmic-ray files: the team's archived Moscow series for the published-value comparison, our NMDB download elsewhere